

## AMAHAI, Indonesia

WMO: 977220

Lat: 3.3464S

Lon: 128.9281E

Elev: 10

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 8             | 21.8       | 22.5 | 20.6                          | 15.3 | 23.7 | 21.1 | 15.8 | 24.2 | 8.2                   | 26.0 | 7.2 | 26.1 | 0.3                   | N/A  | 0.288 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 12            | 6.7                    | 32.5            | 26.3 | 32.0 | 26.3 | 31.5 | 26.2 | 27.5                | 30.6 | 27.0 | 30.5 | 26.7 | 30.1 | 3.0                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.7                            | 22.2 | 29.7 | 26.1 | 21.5 | 29.2 | 25.7 | 21.0 | 28.9 | 87.5          | 30.7 | 85.2 | 30.6 | 83.9 | 30.4 | 31.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)  | (q) |
| (4) 6.0           | 4.9  | 4.2 | DB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | N/A |
| (5)               |      |     | WB                         | 19.9 | 29.2               | 0.9 | 1.4                                                | 19.2 | 30.2       | 18.7 | 31.1       | 18.2 | 31.9       | 17.5 | 32.9 |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 26.8   | 27.9  | 27.9  | 27.8  | 27.4  | 26.9  | 25.9  | 25.2  | 25.1  | 25.7  | 26.7  | 27.6  | 28.0 |
|                                                                        | DBStd       | 1.33   | 0.88  | 0.92  | 0.88  | 0.64  | 0.70  | 0.97  | 0.92  | 0.86  | 0.90  | 0.66  | 0.60  | 0.75 |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                        | CDD10.0     | 6149   | 556   | 501   | 552   | 524   | 478   | 471   | 469   | 472   | 519   | 527   | 559   |      |
|                                                                        | CDD18.3     | 3108   | 297   | 267   | 293   | 272   | 266   | 228   | 213   | 211   | 222   | 261   | 277   | 301  |
|                                                                        | CDH23.3     | 29421  | 3163  | 2851  | 3074  | 2754  | 2555  | 1806  | 1364  | 1356  | 1737  | 2496  | 2949  | 3317 |
| (13)                                                                   | CDH26.7     | 8335   | 1116  | 990   | 1056  | 831   | 666   | 292   | 118   | 117   | 267   | 654   | 1009  | 1218 |
| (14)                                                                   | Wind        | WSAvg  | 1.2   | 1.3   | 1.5   | 1.2   | 1.1   | 1.0   | 1.1   | 1.4   | 1.5   | 1.2   | 1.1   | 1.3  |
| (15) Precipitation                                                     | PrecAvg     | 2586   | 130   | 119   | 171   | 229   | 286   | 371   | 414   | 287   | 200   | 129   | 104   | 145  |
|                                                                        | PrecMax     | 4911   | 287   | 223   | 286   | 461   | 814   | 1086  | 1141  | 1375  | 483   | 373   | 253   | 262  |
|                                                                        | PrecMin     | 1250   | 43    | 39    | 92    | 113   | 60    | 26    | 37    | 0     | 0     | 21    | 27    | 34   |
|                                                                        | PrecStd     | 959    | 59    | 43    | 49    | 77    | 188   | 248   | 262   | 310   | 137   | 91    | 53    | 67   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.8  | 33.2  | 33.3  | 31.6  | 30.8  | 30.0  | 29.0  | 29.1  | 29.8  | 30.8  | 32.0  | 33.2 |
|                                                                        |             | MCWB   | 26.0  | 26.0  | 26.0  | 26.5  | 26.5  | 26.1  | 25.5  | 25.5  | 25.9  | 25.9  | 26.4  | 26.9 |
|                                                                        | 2%          | DB     | 32.1  | 32.2  | 32.1  | 31.0  | 30.4  | 29.5  | 28.4  | 28.5  | 29.3  | 30.3  | 31.5  | 32.3 |
|                                                                        |             | MCWB   | 26.2  | 25.9  | 26.1  | 26.3  | 26.3  | 25.9  | 25.2  | 25.1  | 25.7  | 25.9  | 26.3  | 26.8 |
|                                                                        | 5%          | DB     | 31.6  | 31.5  | 31.3  | 30.6  | 30.0  | 28.9  | 27.9  | 27.9  | 28.7  | 29.9  | 31.0  | 31.7 |
|                                                                        |             | MCWB   | 26.1  | 25.9  | 26.0  | 26.2  | 26.2  | 25.8  | 24.9  | 24.8  | 25.4  | 25.9  | 26.2  | 26.7 |
|                                                                        | 10%         | DB     | 31.0  | 30.9  | 30.7  | 30.2  | 29.6  | 28.4  | 27.4  | 27.3  | 28.2  | 29.5  | 30.6  | 31.2 |
|                                                                        |             | MCWB   | 25.9  | 25.8  | 25.9  | 26.1  | 26.0  | 25.5  | 24.7  | 24.5  | 25.1  | 25.7  | 26.1  | 26.4 |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.2  | 27.2  | 27.2  | 27.4  | 27.1  | 26.7  | 26.1  | 25.9  | 26.4  | 27.0  | 27.7  | 30.0 |
|                                                                        |             | MCDB   | 31.3  | 30.9  | 31.0  | 30.3  | 30.0  | 28.9  | 28.2  | 28.3  | 29.1  | 29.5  | 30.6  | 31.9 |
|                                                                        | 2%          | WB     | 26.7  | 26.7  | 26.7  | 26.8  | 26.7  | 26.2  | 25.6  | 25.5  | 26.0  | 26.5  | 27.1  | 28.3 |
|                                                                        |             | MCDB   | 30.7  | 30.7  | 30.5  | 30.1  | 29.5  | 28.6  | 27.7  | 27.7  | 28.7  | 29.3  | 30.4  | 30.7 |
|                                                                        | 5%          | WB     | 26.5  | 26.4  | 26.4  | 26.6  | 26.5  | 26.0  | 25.2  | 25.1  | 25.7  | 26.2  | 26.7  | 27.0 |
|                                                                        |             | MCDB   | 30.5  | 30.4  | 30.2  | 29.8  | 29.3  | 28.4  | 27.2  | 27.2  | 28.2  | 29.1  | 30.1  | 30.7 |
|                                                                        | 10%         | WB     | 26.2  | 26.1  | 26.1  | 26.3  | 26.2  | 25.7  | 24.9  | 24.7  | 25.3  | 25.9  | 26.4  | 26.6 |
|                                                                        |             | MCDB   | 30.2  | 30.1  | 29.9  | 29.4  | 28.9  | 27.8  | 26.8  | 26.7  | 27.6  | 28.9  | 29.8  | 30.2 |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.5   | 6.4   | 6.5   | 5.8   | 5.3   | 4.4   | 4.0   | 4.0   | 4.8   | 5.7   | 6.4   | 6.7  |
|                                                                        |             | MCDBR  | 7.1   | 6.9   | 6.9   | 6.2   | 5.8   | 5.2   | 4.8   | 4.9   | 5.4   | 6.2   | 6.8   | 7.3  |
|                                                                        |             | MCWBR  | 3.1   | 2.9   | 2.9   | 2.9   | 2.8   | 2.6   | 2.3   | 2.2   | 2.7   | 2.8   | 3.1   | 3.3  |
|                                                                        | 5% WB       | MCDBR  | 6.8   | 6.7   | 6.5   | 6.1   | 5.4   | 4.8   | 4.3   | 4.4   | 5.2   | 5.8   | 6.5   | 6.9  |
|                                                                        |             | MCWBR  | 3.2   | 3.2   | 3.1   | 3.0   | 2.7   | 2.5   | 2.2   | 2.2   | 2.7   | 2.9   | 3.3   | 3.5  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.416  | 0.419 | 0.417 | 0.411 | 0.419 | 0.430 | 0.423 | 0.424 | 0.434 | 0.458 | 0.445 | 0.427 |      |
|                                                                        | taud        | 2.487  | 2.464 | 2.477 | 2.508 | 2.489 | 2.469 | 2.486 | 2.466 | 2.429 | 2.339 | 2.382 | 2.444 |      |
|                                                                        | Ebn at Noon | 919    | 916   | 908   | 890   | 857   | 834   | 846   | 866   | 879   | 871   | 886   | 904   |      |
|                                                                        | Edh at Noon | 115    | 118   | 116   | 108   | 106   | 106   | 106   | 112   | 119   | 132   | 126   | 119   |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 5.11   | 5.34  | 5.36  | 5.11  | 4.67  | 3.93  | 4.02  | 4.65  | 5.09  | 5.74  | 5.68  | 5.22  |      |
| (45)                                                                   | RadStd      | 0.34   | 0.36  | 0.33  | 0.26  | 0.44  | 0.48  | 0.51  | 0.66  | 0.55  | 0.42  | 0.30  | 0.37  |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page